Fiscal Year:	FY 2016	**Task Last Updated:**	FY 04/04/2016
PI Name:	Mueller, Holger Ph.D.		
Project Title:	QWEP Analysis and Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-0001	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kohel, James (Jet Propulsion Laboratory) Yu, Nan (Jet Propulsion Laboratory)		
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The equivalence of gravity and accelerated motion is the basis of general relativity's description of gravity as a result of curved space and time. This description, however, is incompatible with the quantum theories of the electroweak and the strong interactions, in which space and time exist independently of the forces. It is therefore not surprising that unified theories of gravity and quantum mechanics are expected to contain tiny violations of the gravity-acceleration equivalence. Such violations would manifest themselves as variations between the acceleration of free fall for different types of matter. These variations are considered as being among the most promising areas to look for signals arising from a unification of general relativity and quantum mechanics. ESA's (European Space Agency) quantum weak equivalence principle (QWEP) mission seeks such signals with a strongly improved sensitivity of 10-15 or better. By using quantum interference of ultracold atoms, it will use quantum particles where previous tests used bulk matter.

Task Description:	The research proposed here will combine theory and experiment to maximize the sensitivity and scientific impact of QWEP, and minimize the risk. Using the standard model extension (SME) as a theoretical framework, we will study which species of atoms, what quantum states, and what type of analysis might lead to the largest and most interesting signals. The choice of species, however, is equally dependent on experimental considerations. Using our existing atom interferometers, we will experimentally compare different technologies for the matter-wave beam splitters that are suitable for the candidate species, and develop strategies to minimize systematic effects such as rotation and gravity gradients. Based on the experience of Jet Propulsion Laboratory (JPL) and the Space Sciences Laboratory (SSL) at University of California (UC) Berkeley, we will match these technologies to the requirement of space operation. Thereby, this work will apply fundamental theory, atomic physics experiment, and principles of astronautics to a comprehensive study optimization and analysis of the QWEP mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We seek to understand two things: What is the difference between a quantum and classical test of the weak equivalence principle? What kind of quantum test would provide the most new information regarding Einstein's principle of equivalence? We have made progress on answering these questions, which have been pursued both theoretically and experimentally.
Task Progress:	In the one-year period directly preceding 6/28/2016, we were focusing on studying the physics background of testing the equivalence principle. So far, both general relativity and quantum mechanics have been successful theories, as recently demonstrated by the spectacular detection of gravitational waves. General relativity follows from the Einstein equivalence principle, which holds that gravity is indistinguishable from the effects of being in an accelerated reference frame by any experiment whatsoever. General relativity and quantum mechanics are believed to be the low-energy limits of an unknown fundamental theory. While the energy scale of these fundamental theories is typically inaccessible to experiments, it is possible to search for suppressed effects at accessible energy scales in experiments of outstanding sensitivity. Thus, fundamental physics can be tested by looking for violations of equivalence principle at experimentally accessible scales. The form of such low-energy effects, however, remains unknown. Thus, it is important to search for EEP violation in as many different places as possible. A well-known method for such searches compares the classical motion of test particles using, e.g., torsion balances and lunar laser ranging. It has reached an accuracy of ~10-13 for the ratio of the acceleration of free fall between a few species, known as the Eötvös parameter. But the equivalence principle is not restricted to classical observables, and violations not ruled out by the classical tests could become apparent when the quantum degrees of freedom are controlled or observable, such as spin or the phase of a wave function. Quantum tests offer distinctive features compared to classical tests. The coherence length of quantum objects is several orders of magnitude larger than the classical counter parts. Redshift and UFF tests can be performed with the same species. Quantum tests typically utilize spin-polarized ensembles which are hardly accessible by other techniques. Additionally, atom optic techniques prepare samples of pure isotopes implying well defined test bodies. Finally, quantum tests give access to complementary test bodies which would be difficult to machine for a classical apparatus. Another reason is exploring modified gravity theories that are models of dark energy. Dark energy could be a dynamic field - quintessence. The original and simplest model is the so-called tracker potential. Confirming this model would identify the dominant constituent of the universe, help understanding its history, and be a milestone in particle physics. In principle, this is possible by studying the history of the dark energy density, e.g., by improved supernova surveys, microwave-background-radiation, or structure-formation studies. While these measurements are important, no single approach is guaranteed to identify the quintessence field. Making the reasonable assumption of an additional coupling between dark energy and normal matter, of roughly gravitational strength, allows us to detect quintessence by measuring its interaction with cold atoms in atom interferometers. We have already constrained the model more than 1000 times more strongly than the best previous experiment. Searching the entire range of allowed interaction strengths is feasible. We would then either detect the field or rule out its existence (always assuming that quintessence does indeed interact -- but showing that it doesn't would be interesting in its own right if quintessence is confirmed by cosmological data, because it would characterize its properties). Experiments in space can accomplish this because they offer long interaction times of an atomic cloud with a source mass. Finally, measurements of gravity will benefit particularly strongly from microgravity. The gravitational constant G is the least well-known fundamental constant: According to the latest (2014) CODATA adjustment, G = 6.67408(31) ×10-11 m3kg-1s-2 with a relative uncertainty of 47 ppm, but individual measurements can disagree by several times their stated error bar. This is a clear hint that the systematic errors of such measurements (mostly torsion balances) are not well-understood, making independent measurements with a different set of systematic highly desirable. One possibility is atom interferometers, which have recently achieved an error of 140 ppm. This total error arises mostly from position uncertainty of the atoms relative to the masses, and the homogeneity of the masses. An atom interferometry mission in microgravity can park the atoms closely to the field-generating masses, so that masses of 100s of grams rather than 100s of kilograms will be sufficient to generate a strong signal. The small source masses can be made of high-purity (99.99-99.999%) single crystals of tungsten or tantalum. This will suppress errors due to inhomogeneity and impurity of the masses. A new class of measurements of G made possible by space operation is provided by the gravitational Aharonov-Bohm effect. Using a pair of test masses creates a W-shaped gravitational potential. Placing the atomic wave packet in a superposition of being in a maximum and a minimum (more accurately, saddle points) of the potential will allow us to measure a phase shift caused by a gravitational potential difference. This is not possible on the ground, where gravity will pull the atoms away after milliseconds.
Bibliography Type:	Description: (Last Updated: 01/29/2019)

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